Business Performance Analysis Report

Superstore Sales Dataset (2014–2017)

Executive Summary

This report analyzes four years of sales transactions (2014–2017) to evaluate overall business performance, regional and category-level trends, the impact of discounting on profitability, and customer segment contributions. The dataset recorded total sales of approximately $9.19 million and total profit of approximately $572,794, an overall profit margin of around 6.2%.

1. Is the Company Growing Month-over-Month?

The month-over-month growth analysis (calculated using SQL window-style self-joins on monthly sales) shows that growth is highly volatile rather than steadily increasing. Several months show extreme swings — for example, August 2016 sales grew by over 3,000% compared to July 2016, while several other months show declines exceeding 80–90% (e.g., December 2014 at -94.7%, October 2014 at -94.3%, and September 2016 at -92.4%).

Looking at the broader trend across 2017 (the most recent full year in the data), monthly sales fluctuate between roughly $39,000 and $89,000, with no clear sustained upward trajectory. Overall, the company shows seasonal spikes (notably around August–September and November–December, consistent with back-to-school and holiday shopping) but does not demonstrate consistent month-over-month growth. Growth appears to be driven by seasonality and large individual orders rather than steady organic expansion.

2. Which Region is Underperforming?

Profitability by region (SQL Query 4) reveals two regions operating at a net loss:

Region	Total Profit ($)
South	-134.40
East	-1,071.85
Central	173.89
West	1,194.59

The East region is the most significantly underperforming, posting a loss of over $1,070 despite generating sales — indicating costs, discounting, or returns are eroding margins disproportionately. The South region is also unprofitable overall, though the loss is comparatively small. In contrast, the West region is the strongest performer, followed by

Central. Management should prioritize a profitability investigation in the East region first, as it represents the largest negative impact.

3. Which Category Has the Highest Growth?

Comparing total sales and profit across product categories:

Category	Total Sales ($)	Total Profit ($)	Profit Margin
Technology	836,154.03	145,454.95	17.4%
Office Supplies	719,047.03	122,490.80	17.0%
Furniture	741,999.80	18,451.27	2.5%

Technology is the strongest category overall, generating the highest sales and the highest profit, with a healthy ~17% margin. Office Supplies performs similarly well on margin despite slightly lower sales. Furniture, however, stands out negatively: although its sales volume is comparable to the other two categories, its profit margin is drastically lower at just 2.5%, suggesting heavy discounting or low-margin product lines (such as tables and bookcases) are dragging down profitability. Technology can be considered the category with the strongest growth and profitability profile, while Furniture is the weakest and a key area for improvement.

4. Are Discounts Reducing Profit?

Yes. The data shows a clear negative relationship between discount level and profit (correlation coefficient of approximately -0.22 between Discount and Profit). The average profit per order at different discount levels is as follows:

Discount Level	Average Profit per Order ($)
0% (No Discount)	66.90
10%	96.06
15%	27.29
20%	24.70
30%	-45.68
40%	-111.93
50%	-310.70
70%	-95.87
80%	-101.80

The pattern is unambiguous: orders with discounts of 0–20% remain profitable on average, while any discount of 30% or higher results in average losses, with the loss deepening sharply as the discount rate increases (peaking around -$310 average loss at 50% discount). This strongly suggests that deep discounting practices — particularly discounts of 30% and above — are a major driver of unprofitable orders and should be tightly controlled.

5. Which Customer Segment Contributes Most Revenue?

Segment	Total Sales ($)	Total Profit ($)
Consumer	1,161,401.00	134,119.21
Corporate	706,146.40	91,979.13
Home Office	429,653.10	60,298.68

The Consumer segment contributes the most revenue by a wide margin, accounting for roughly 49% of total sales and 47% of total profit, more than the Corporate and Home Office segments combined contribute individually. Corporate is the second-largest contributor, while Home Office, though the smallest, still maintains a healthy profit-to-sales ratio. All three segments are profitable, but the business is heavily dependent on the Consumer segment for both revenue and profit generation.

6. What Strategic Actions Should Management Take?

- Cap discounts at 20%: Since average profit turns negative beyond a 20% discount and losses grow severely past 30%, management should establish discount approval thresholds and require managerial sign-off for any discount above 20%.
- Investigate and turn around the East and South regions: Conduct a deep-dive into the East region's cost structure, shipping expenses, and discount practices, as it is the largest source of regional losses. Apply lessons from the high-performing West region.
- Improve Furniture category margins: Review pricing, supplier costs, and discount frequency for Furniture sub-categories (especially Tables and Bookcases), which appear to be sold at high discounts and low margins relative to Technology and Office Supplies.
- Double down on Technology and Office Supplies: These categories deliver the best margins; management should consider expanding marketing spend, inventory, and promotions for these high-performing categories.
- Leverage the Consumer segment further: Since Consumer drives the most revenue and profit, targeted loyalty programs, personalized promotions, and retention campaigns for this segment could yield strong returns. At the same time, explore growth opportunities in the Home Office segment, which is currently underutilized.

- Smooth seasonal volatility: Given the highly volatile month-over-month growth pattern, management should plan inventory, staffing, and cash flow around known seasonal peaks (August–September and November–December) and design promotions to stimulate sales during historically weaker months.
- Implement ongoing profitability monitoring: Use a recurring (e.g., monthly) dashboard — such as the Power BI report developed alongside this analysis — to track sales, profit, discount impact, and regional/category performance, enabling faster corrective action.

Conclusion

While the business generates substantial overall revenue and remains profitable in aggregate, performance is uneven across regions, categories, and discount tiers. The most pressing issues are excessive discounting (especially at 30%+ levels), underperformance in the East and South regions, and weak margins in the Furniture category. Addressing these areas — while continuing to invest in the strong-performing Technology, Office Supplies, and Consumer segment — represents the clearest path to improved and more consistent profitability.